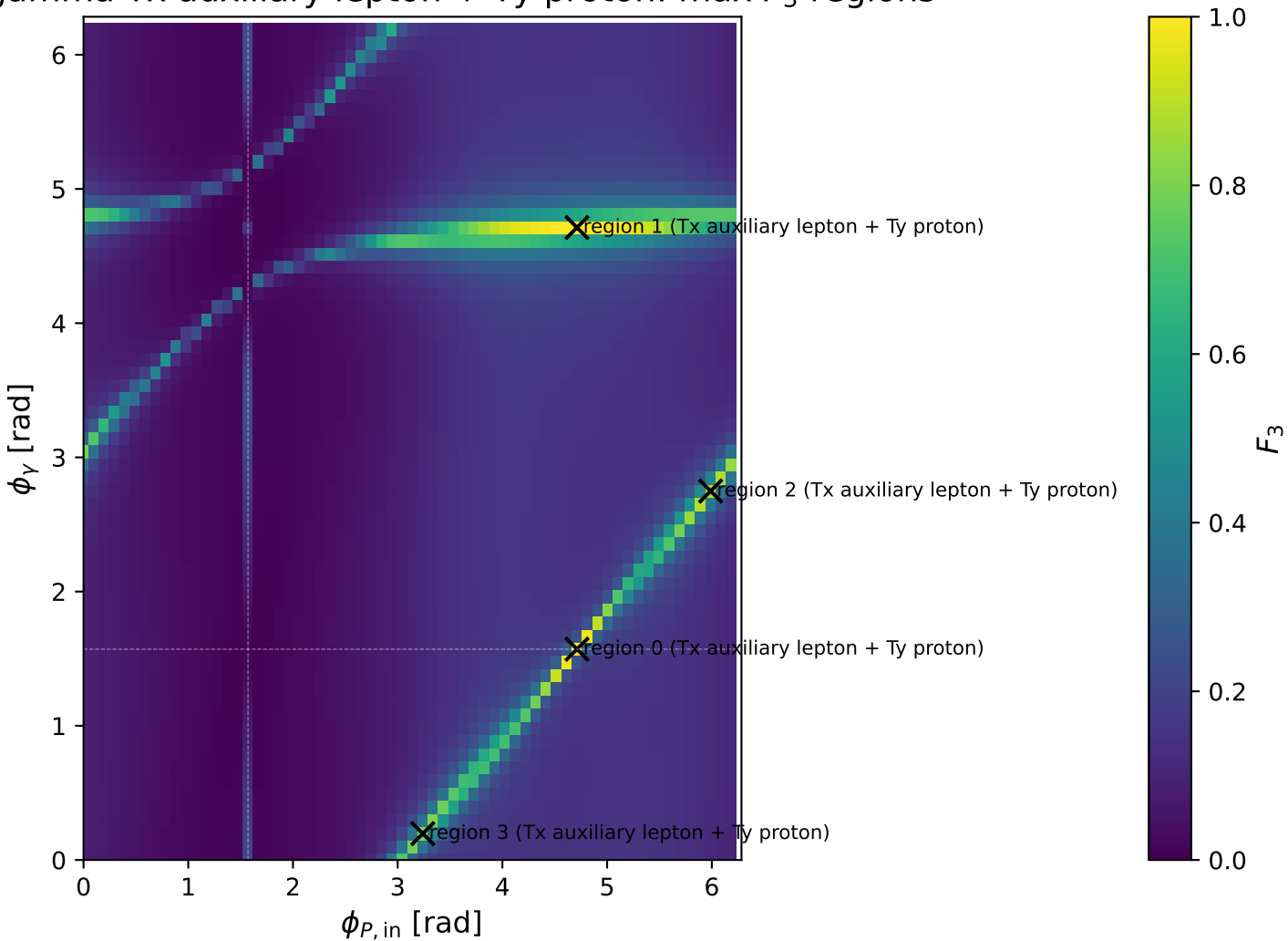
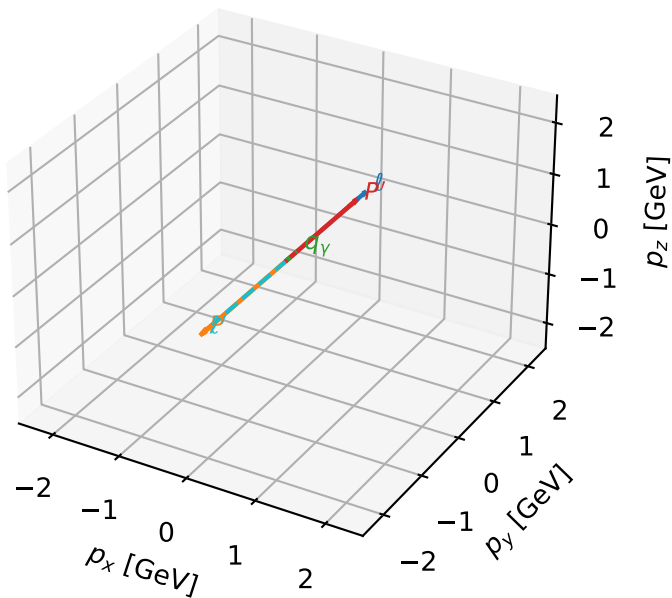


gamma Tx auxiliary lepton + Ty proton: max F_3 regions



Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

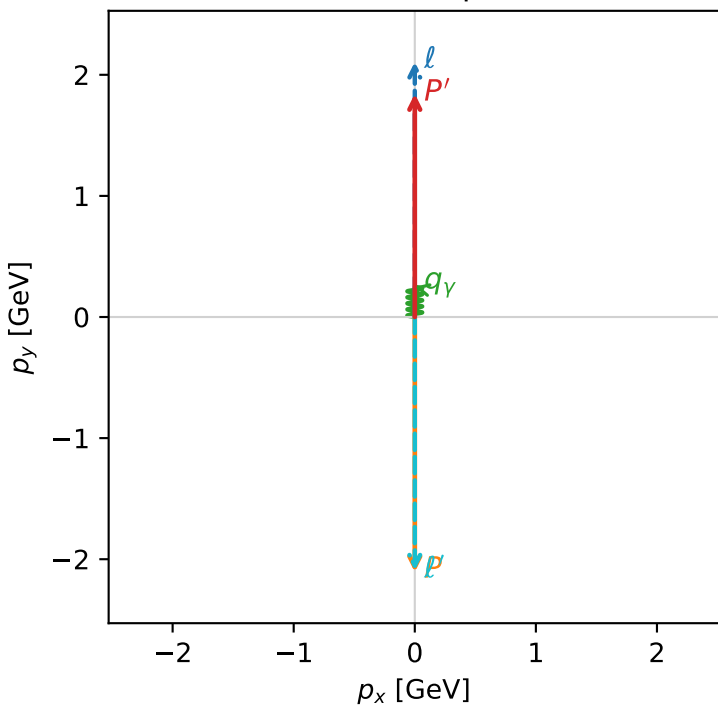


f3_low_egamma_txtty_region_0 $F_3=0.996521$
 region: low_Egamma_TxTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.84352$
 $\theta_{in}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_\gamma=0.25$, $\phi_\gamma=1.5708$
 $Q^2=17.6437$, $x_B=0.993667$, $t=-15.5495$, $W^2=0.9923$, $y=0.904266$
 $\theta(l',\gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= 0.0000$
 l' $E= 2.3201$ $p_x= -0.0000$ $p_y= -2.0935$ $p_z= 0.0000$
 P' $E= 2.0684$ $p_x= 0.0000$ $p_y= 1.8435$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.0000$ $p_y= 0.2500$ $p_z= 0.0000$

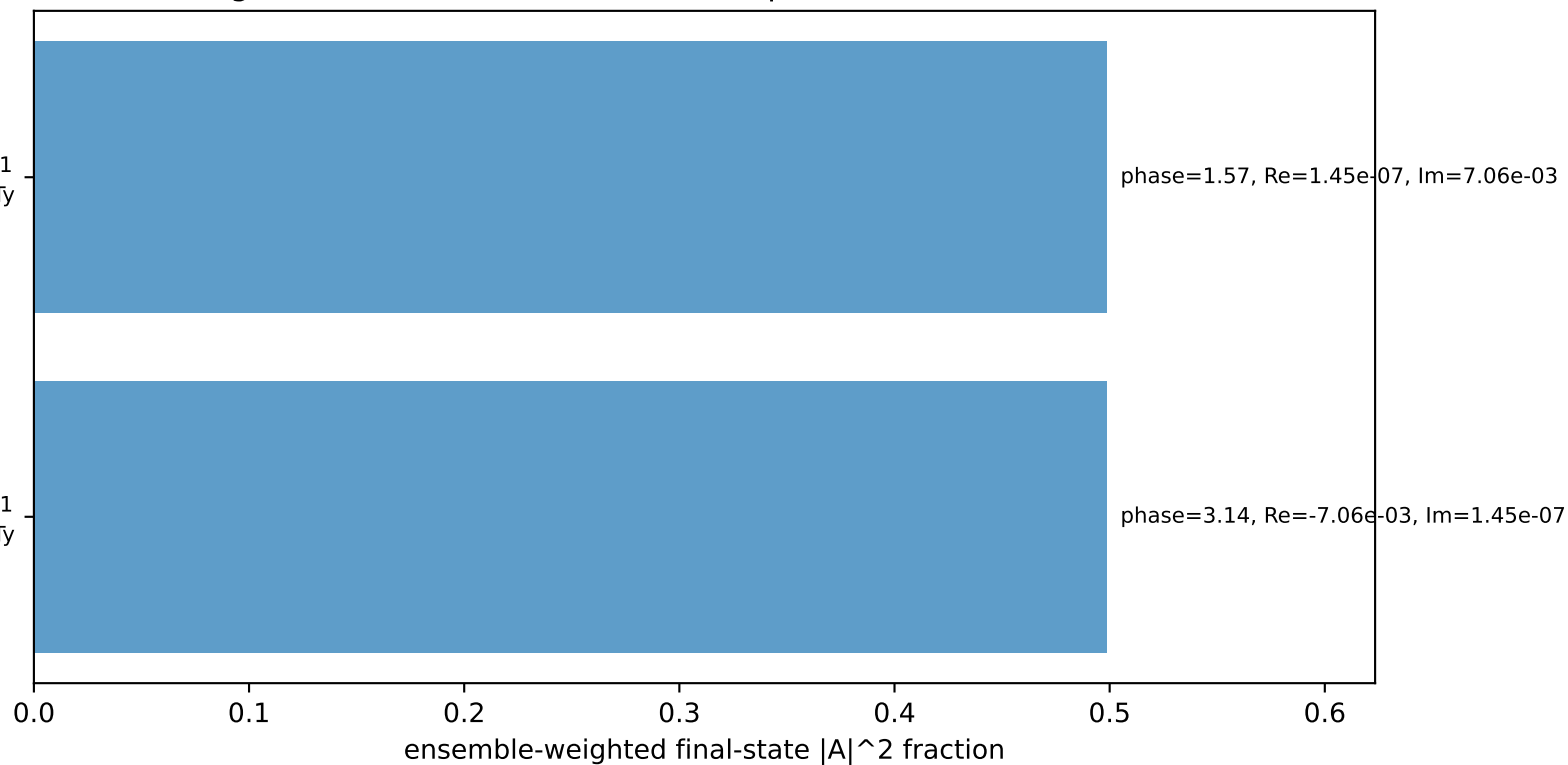
Transverse plane



$h_e=+1$, $h_p=+1$, $h_\gamma=-1$
 electron Tx, proton Ty

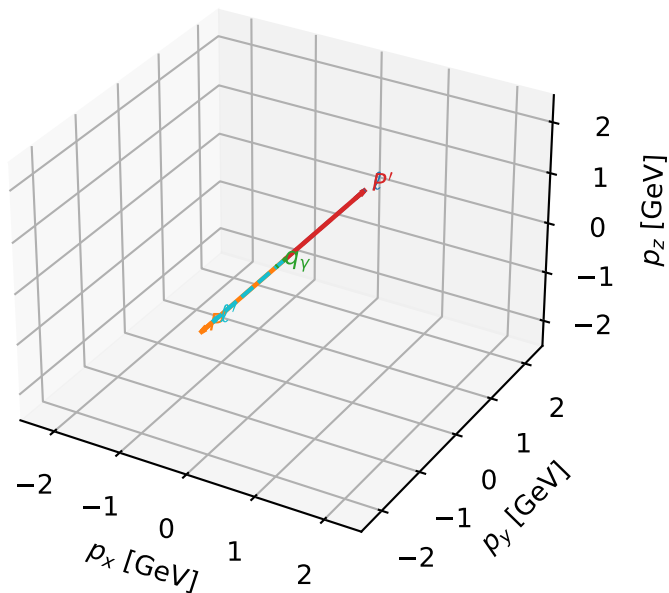
$h_e=-1$, $h_p=-1$, $h_\gamma=+1$
 electron Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=99.7%)



Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

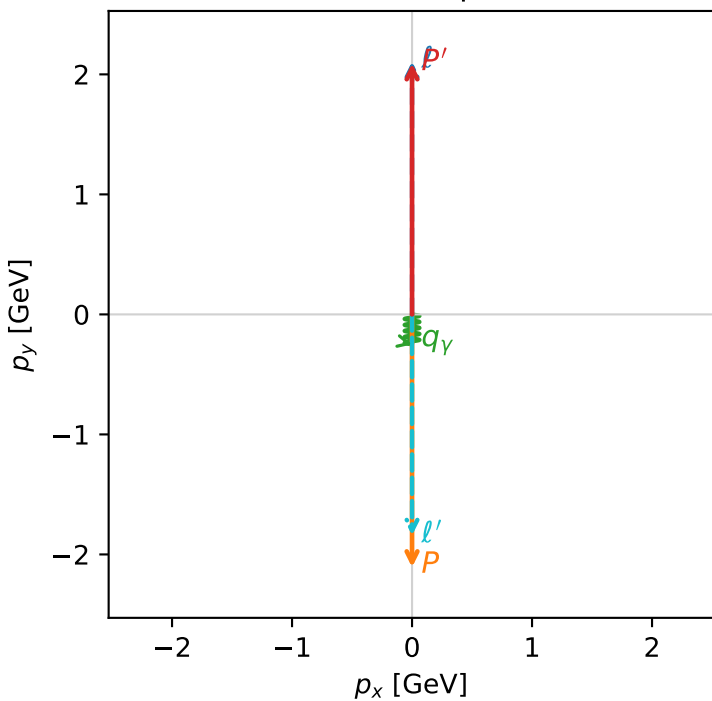


f3_low_egamma_txtty_region_1 F_3=0.995627
 region: low_Egamma_TxTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.09195$
 $\theta_{in}=1.57079$, $\phi_\ell=1.5708$, $\phi_p=4.71239$
 $E_\gamma=0.25$, $\phi_\gamma=4.71239$
 $Q^2=15.5379$, $x_B=0.876353$, $t=-17.6305$, $W^2=3.07213$, $y=0.902944$
 $\theta(\ell', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= 0.0000$
 ℓ' $E= 2.0959$ $p_x= 0.0000$ $p_y= -1.8420$ $p_z= 0.0000$
 P' $E= 2.2926$ $p_x= 0.0000$ $p_y= 2.0920$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.0000$ $p_y= -0.2500$ $p_z= 0.0000$

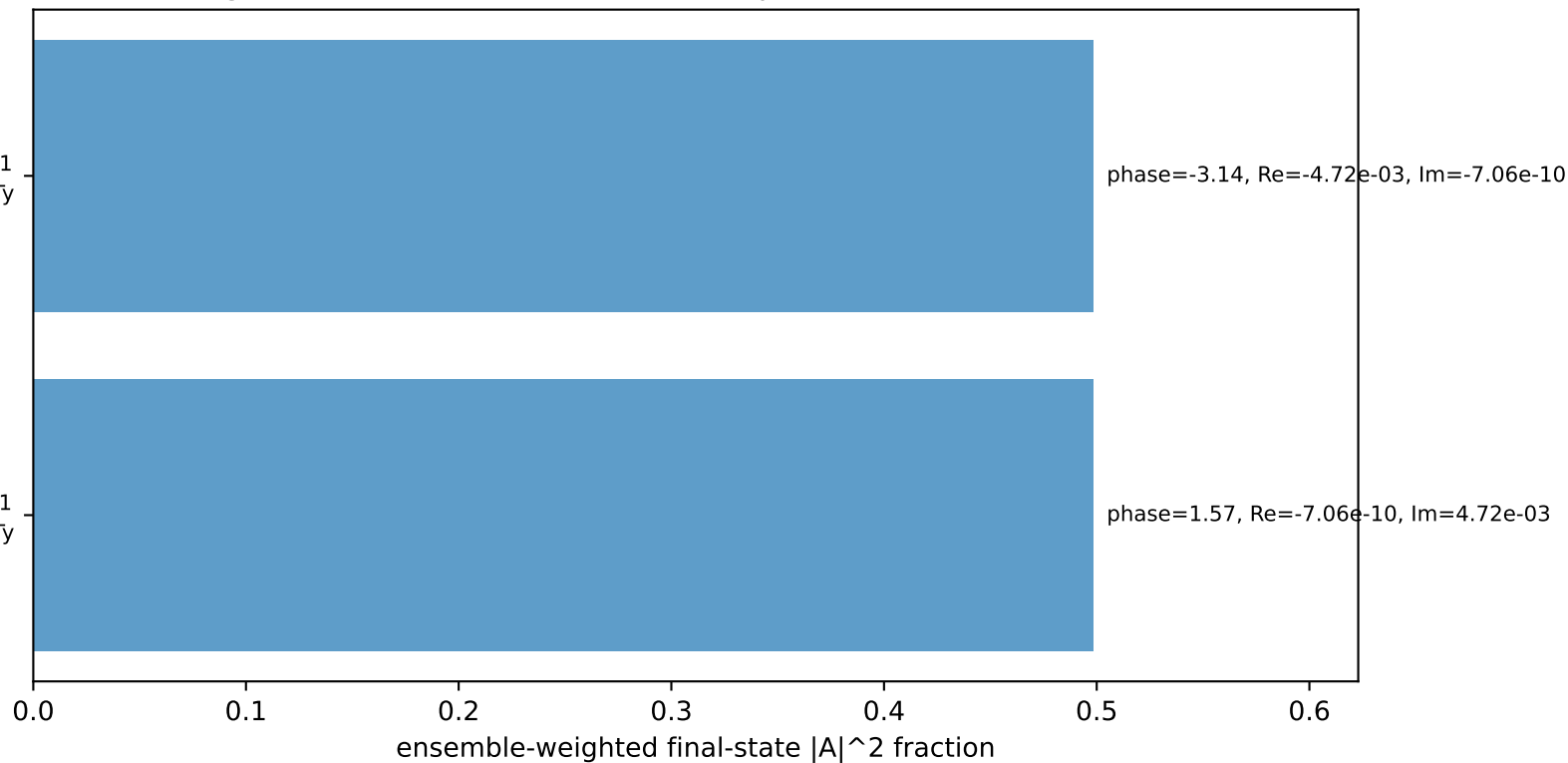
Transverse plane



$h_e=+1$, $h_p=-1$, $h_\gamma=-1$
 electron Tx, proton Ty

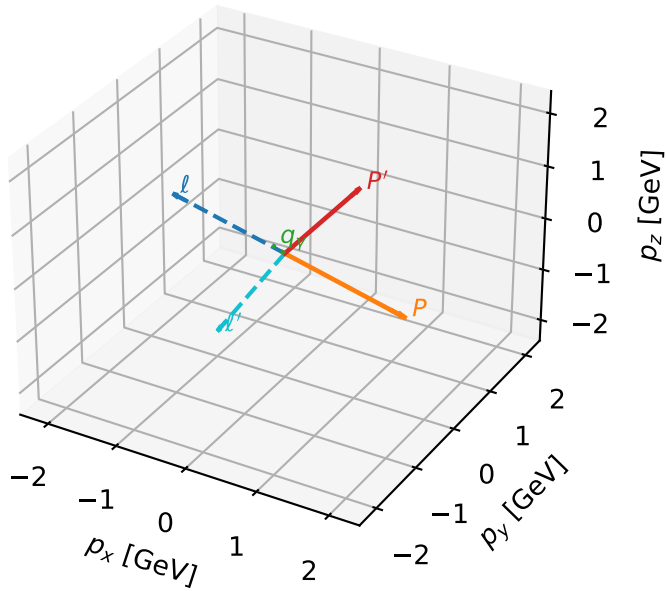
$h_e=-1$, $h_p=+1$, $h_\gamma=+1$
 electron Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=99.7%)



Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

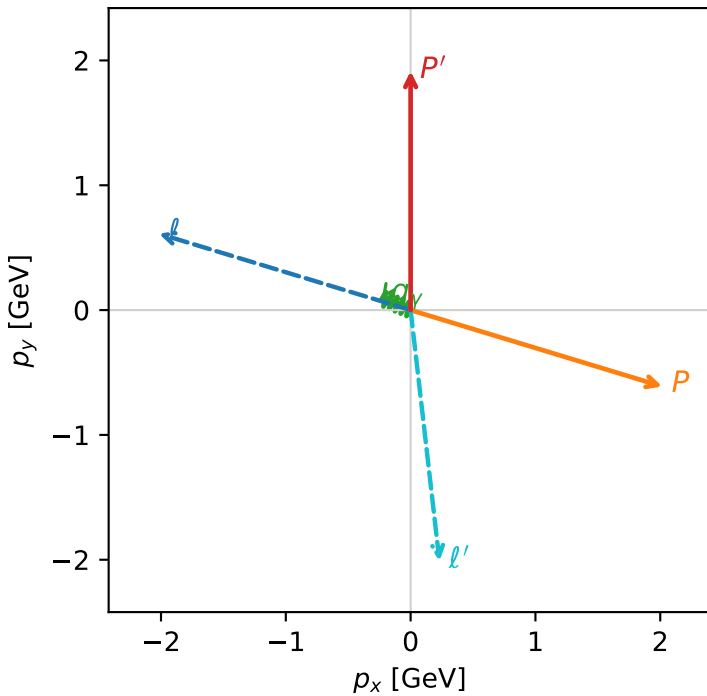


f3_low_egamma_txtty_region_2 F_3=0.896875
 region: low_Egamma_TxTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |\bar{T}_{y,p}\rangle$

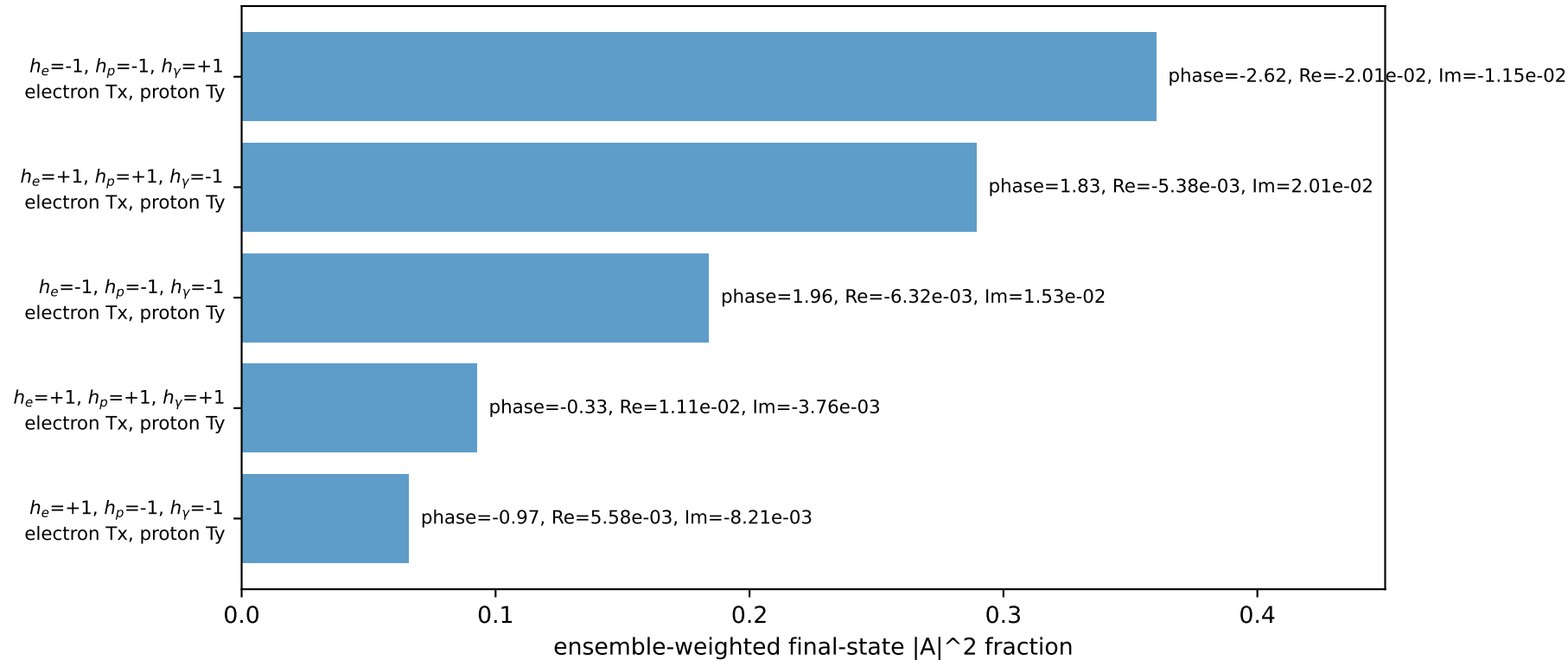
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.91426$
 $\theta_{in}=1.57079$, $\phi_l=2.84707$, $\phi_p=5.98866$
 $E_\gamma=0.25$, $\phi_\gamma=2.74889$
 $Q^2=11.9166$, $x_B=0.944549$, $t=-10.4147$, $W^2=1.57943$, $y=0.642504$
 $\theta(l', \gamma)=119.055$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -2.0162$ $p_y= 0.6116$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 2.0162$ $p_y= -0.6116$ $p_z= 0.0000$
 l' $E= 2.2568$ $p_x= 0.2310$ $p_y= -2.0099$ $p_z= 0.0000$
 P' $E= 2.1317$ $p_x= 0.0000$ $p_y= 1.9143$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2310$ $p_y= 0.0957$ $p_z= 0.0000$

Transverse plane

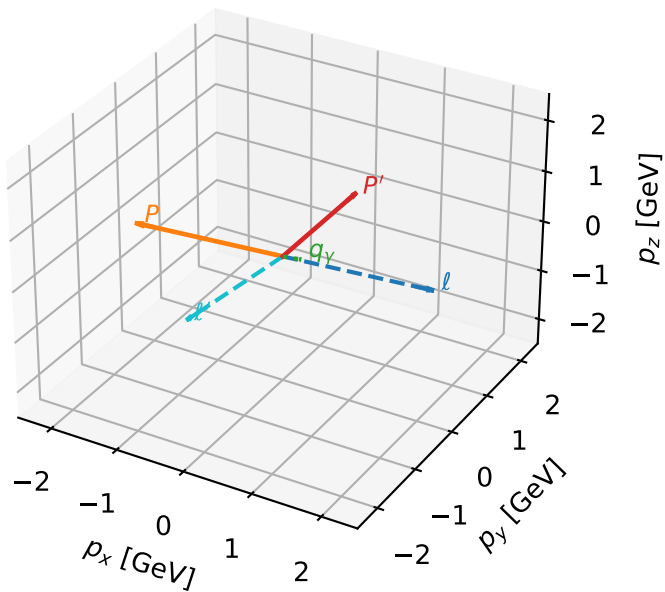


Leading initial-ensemble/final-state components (N=5, retained=99.3%)



Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

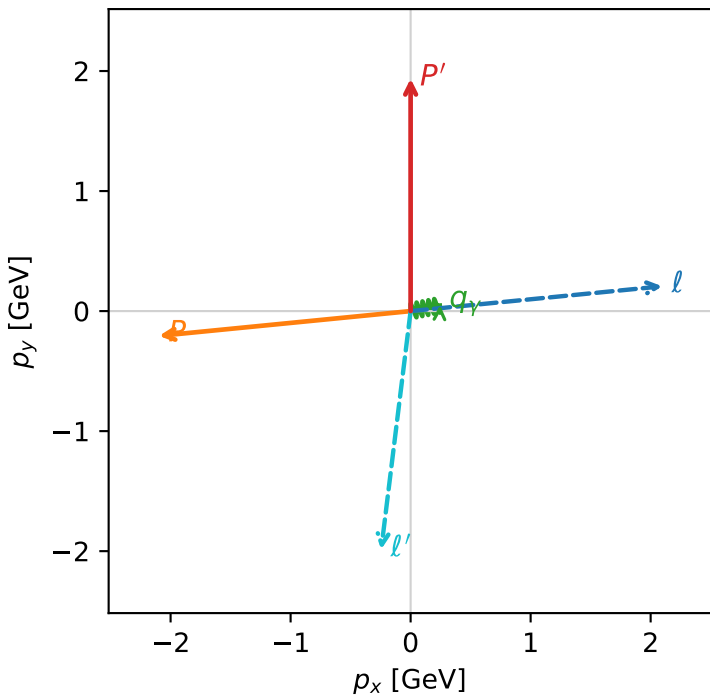


f3_low_egamma_txtty_region_3 F_3=0.811565
 region: low_Egamma_TxTy_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.93673$
 $\theta_{in}=1.57079$, $\phi_l=0.0981748$, $\phi_p=3.23977$
 $E_Y=0.25$, $\phi_Y=0.19635$
 $Q^2=10.2807$, $x_B=0.920571$, $t=-8.96624$, $W^2=1.76689$, $y=0.568741$
 $\theta(l', \gamma)=108.29$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 2.0968$ $p_y= 0.2065$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -2.0968$ $p_y= -0.2065$ $p_z= 0.0000$
 l' $E= 2.2366$ $p_x= -0.2452$ $p_y= -1.9855$ $p_z= 0.0000$
 P' $E= 2.1519$ $p_x= 0.0000$ $p_y= 1.9367$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.2452$ $p_y= 0.0488$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=5, retained=99.4%)

